

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	Hard

Question

K.D. Leka and colleagues found that the Sun’s corona provides an advance indication of solar flares—intense eruptions of electromagnetic radiation that emanate from active regions in the Sun’s photosphere and can interfere with telecommunications on Earth. Preceding a flare, the corona temporarily exhibits increased brightness above the region where the flare is _____.

Which choice completes the text with the most logical and precise word or phrase?

Answer

- A. antecedent
- B. impending
- C. innocuous
- D. perpetual

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of advance indications of solar flares. In this context the word "impending" means imminent or approaching. The text mentions a study by Leka and colleagues that found that the Sun’s corona provides an advance indication of solar flares. The text then points out why such an advance indication would be useful—solar flares can interfere with communications on Earth—and concludes by describing the characteristic of the corona that gives warning of a solar flare. The text indicates that this characteristic—increased brightness in a particular region of the corona—comes before the appearance of the flare. Therefore, in context, the best answer would indicate that the flare is approaching, or impending.

Choice A is incorrect. The best answer would be one that indicates that the increased brightness of the Sun’s corona precedes the appearance of the flare. But if the flare were "antecedent," or previous, then the flare would instead precede the appearance of the increased brightness of the corona, a statement that is logically inconsistent. Choice C is incorrect. The word "innocuous," or harmless, does not logically complete the text; since solar flares can interfere with communications on Earth, they cannot reasonably be described as innocuous. Choice D is incorrect. If the solar flares have an advance indication of their appearance, then there must therefore be a time before the appearance of the flares when they do not exist. But the word "perpetual," or never-ending, would in context indicate that the flare exists at the same time as the advance indication provided by the Sun’s corona, which would not make logical sense.